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Traffic Grooming in OTN2

Comparing the performance of grooming at the original service-rate and at the lower granularity

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CND final project in collaboration with
Nokia company

2024-2025

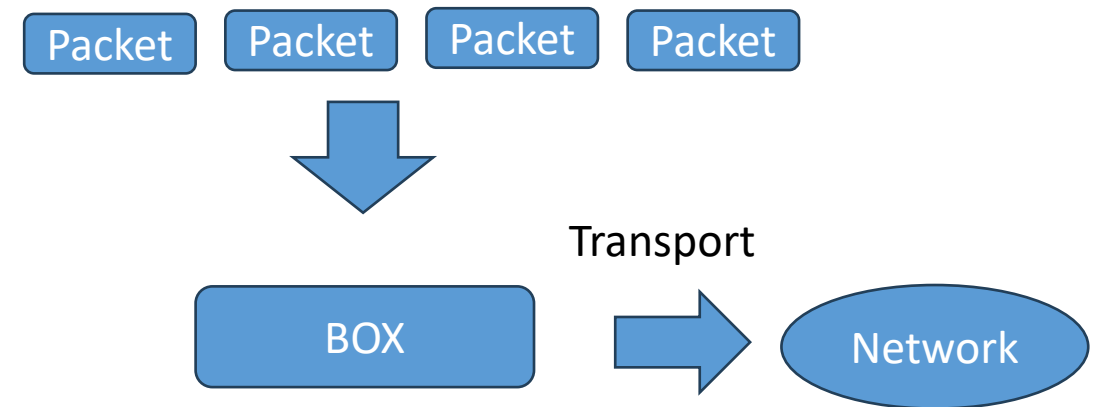
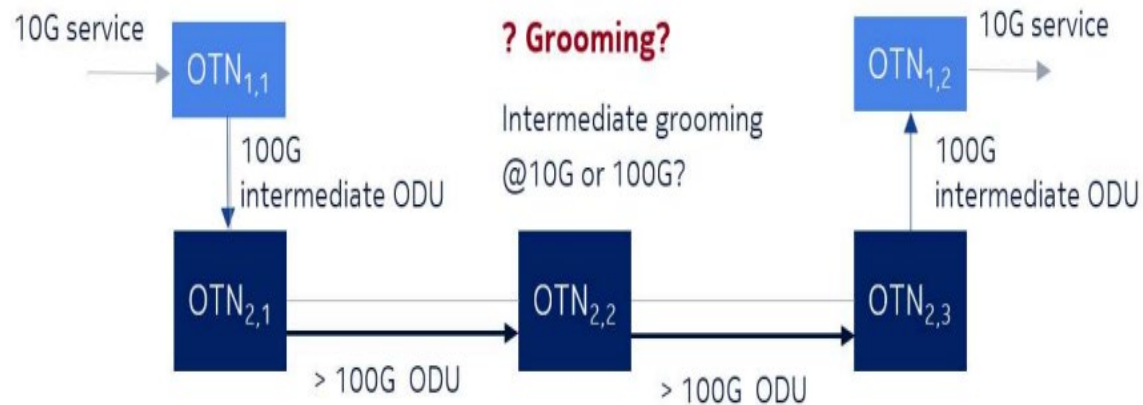
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Traffic Grooming

The process of aggregating multiple low-rate services into a single high-capacity optical lightpath to save spectrum and reduce the number of costly optical transponder.

Cascading Effect

An optical Transport Network (OTN) service can traverse multiple OTN switches before reaching the wavelength Division Multiplexing (WDM) network. It considers a scenario with two OTN switching levels.



Problem Statement

Given

- Network topology
- Set of traffic requests (traffic matrix)
- Set of hierarchical OTN grooming boards and interfaces

Decide

- Original Service-Rate Grooming : Each OTN switch processes at its initial granularity.
- Intermediate Granularity Grooming : Traffic is processed at an intermediate rate that matches the format of the previous OTN switch's output.

Objective

It Compares the performance of grooming at different granularities and analyze the trade-offs between:

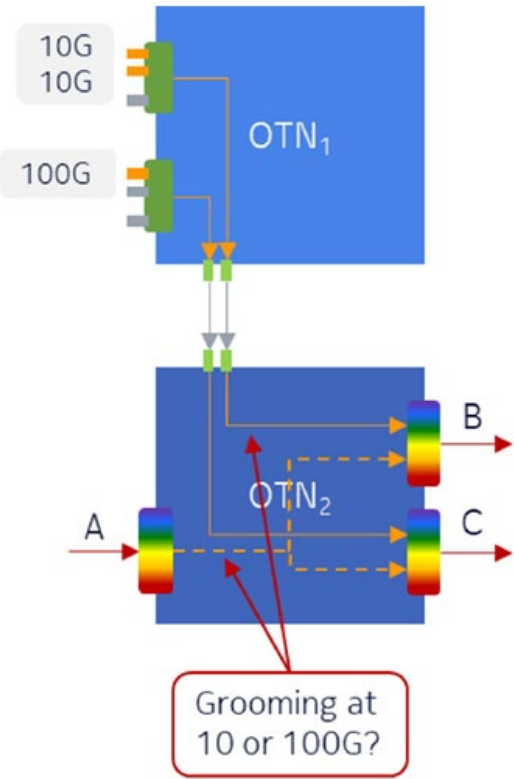
- OTN Matrix Capacity: Considering the effect of different granulation if services on capacity and ODU's OTN1 & OTN2.
- Fiber Saturation: How much of the fiber's capacity is utilized, potentially affecting network congestion and scalability.

Constraints

- Traffic-processing capacity of OTN boards and interfaces
- Maximum number of client interfaces per board
- Wavelength capacity limits

Example

- Considering the effect of different granulation, if services on capacity and ODU's of OTN1 & OTN 2



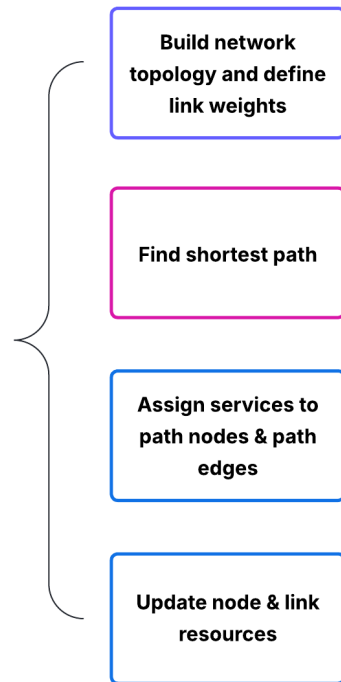
OTN2	Used capacity, Gb/s	Occupied capacity, Gb/s	Number of used ports
OTN1-OTN2 I/O cards	2*100	200	2
Optical transponders	3*200	1500	3

OTN1	Used capacity, Gb/s	Occupied Capacity, Gb/s	Number of used ports
Client I/O cards	2*10	100	1
	1*100	100	1
OTN1-OTN2 I/O cards	2*10 1*100	200	2

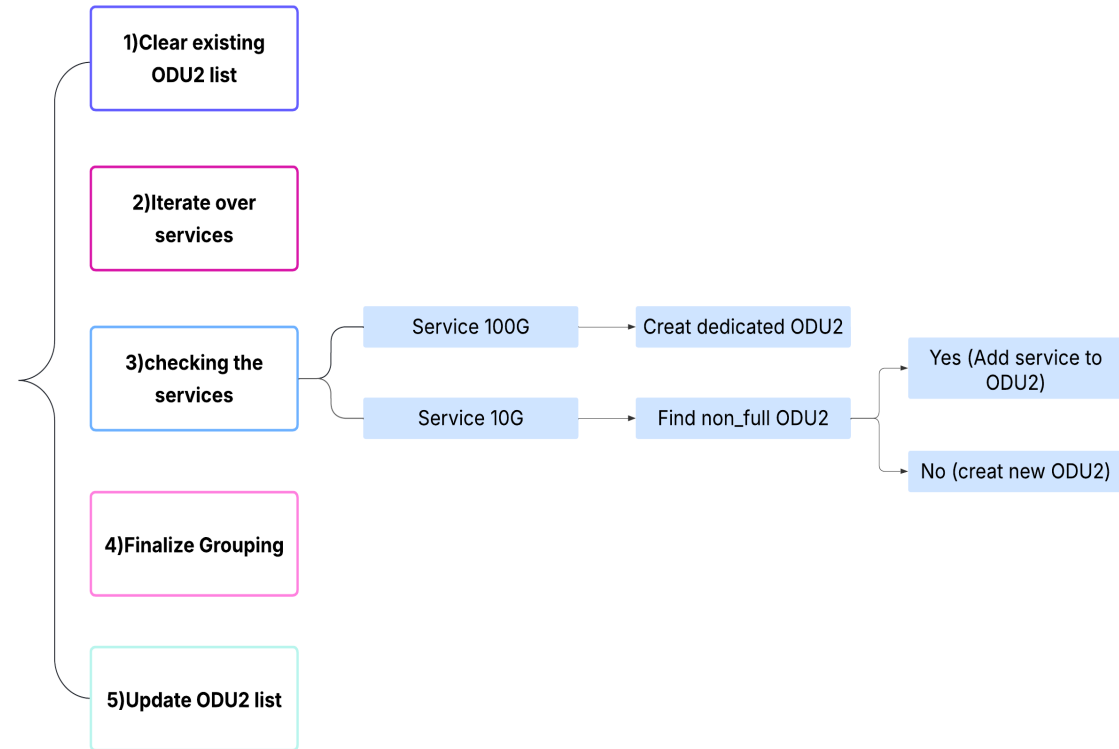
	10G grooming	100G grooming
OTN1	3	3
OTN2	7	4



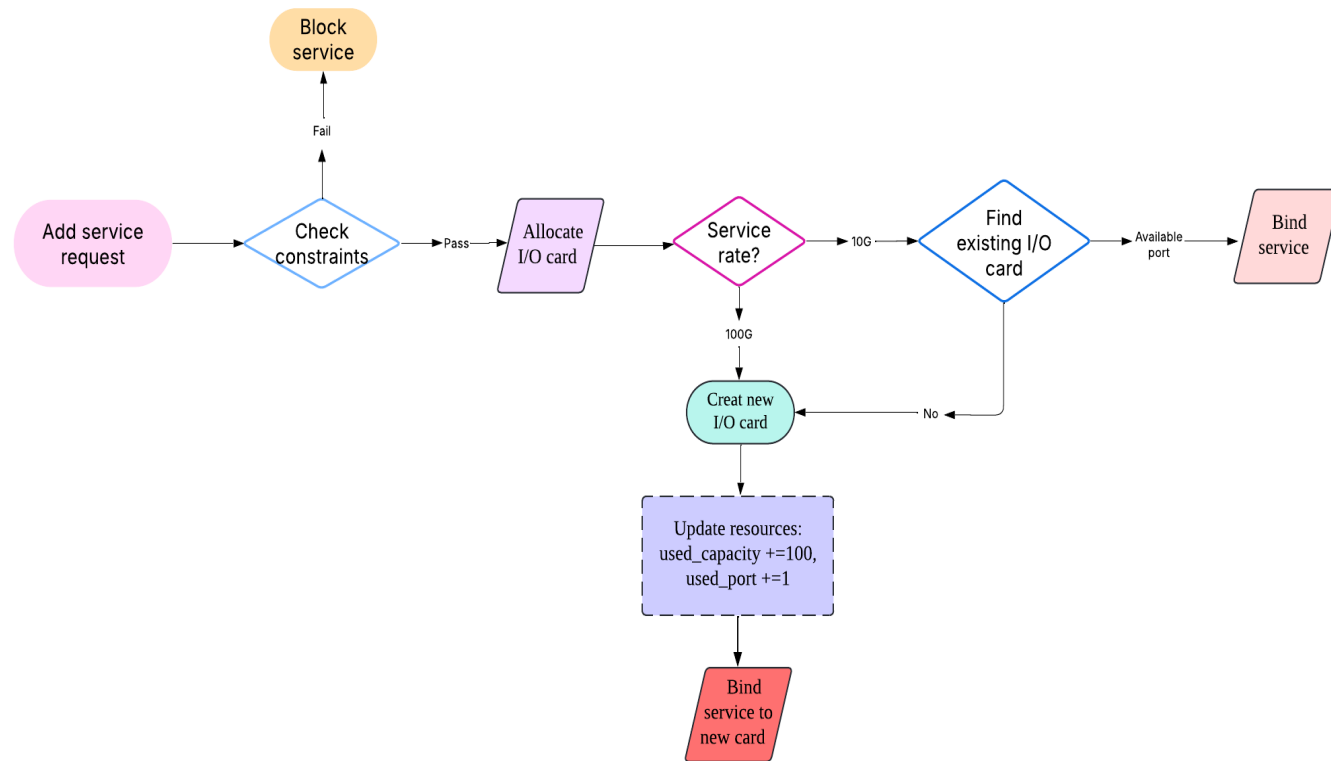
Heuristic approach for routing services



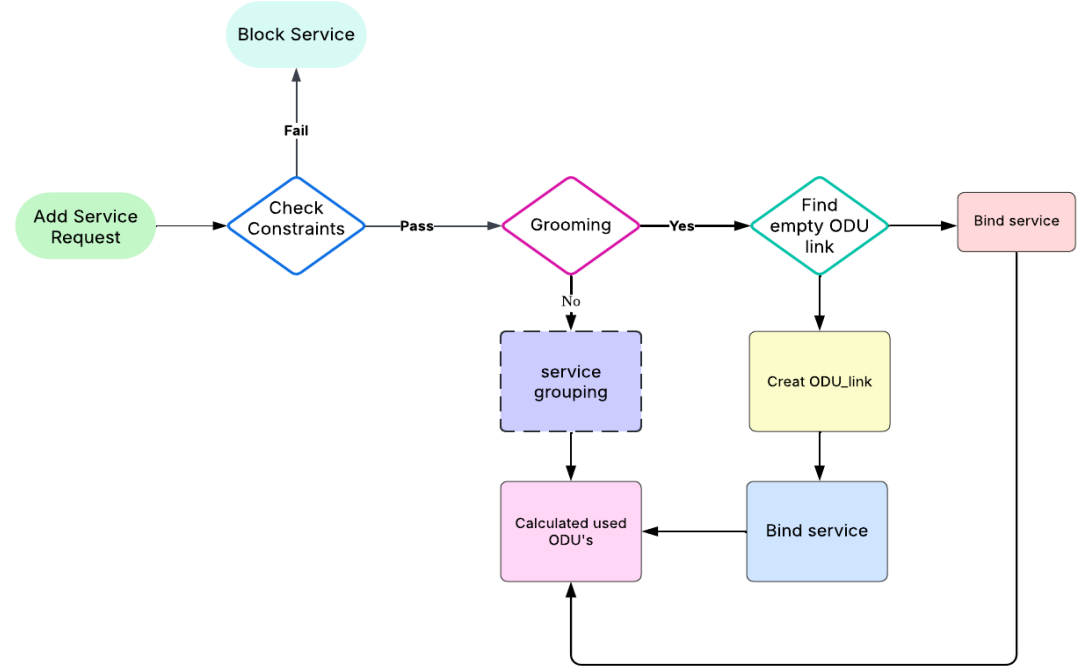
Grooming Process



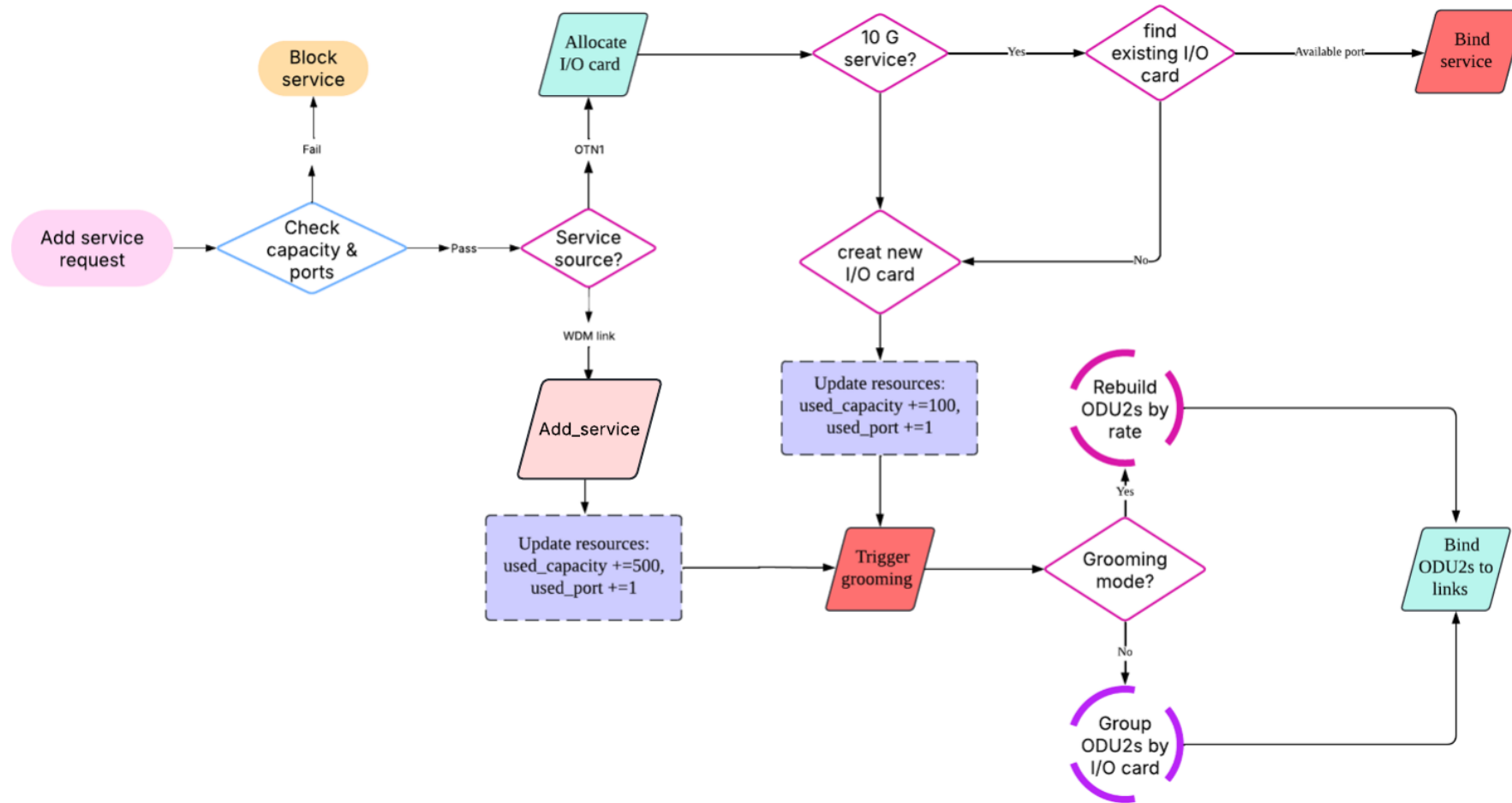
Node (OTN1 & OTN2 switches)



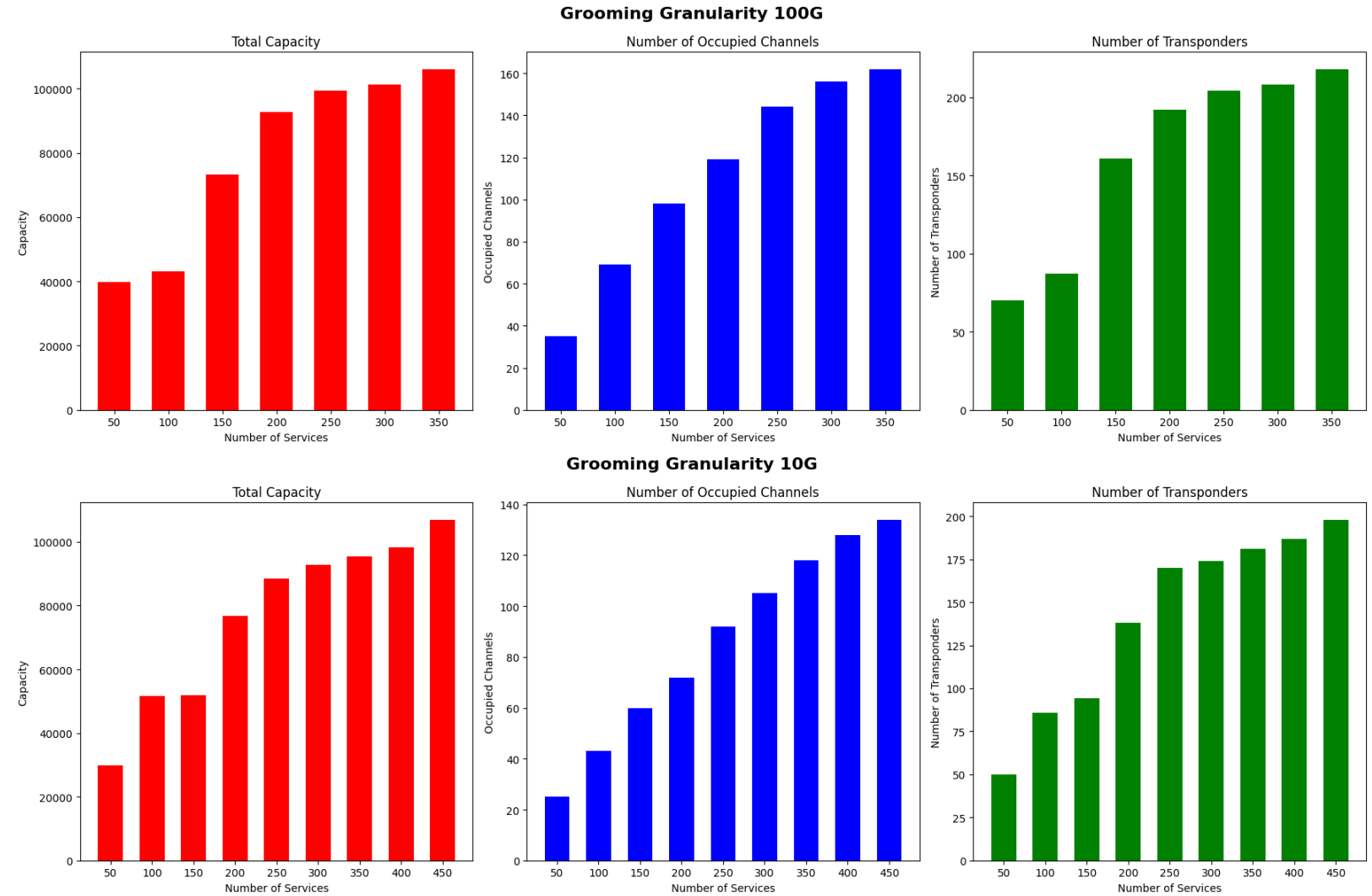
Add server to links



Model of OTN2 Processing

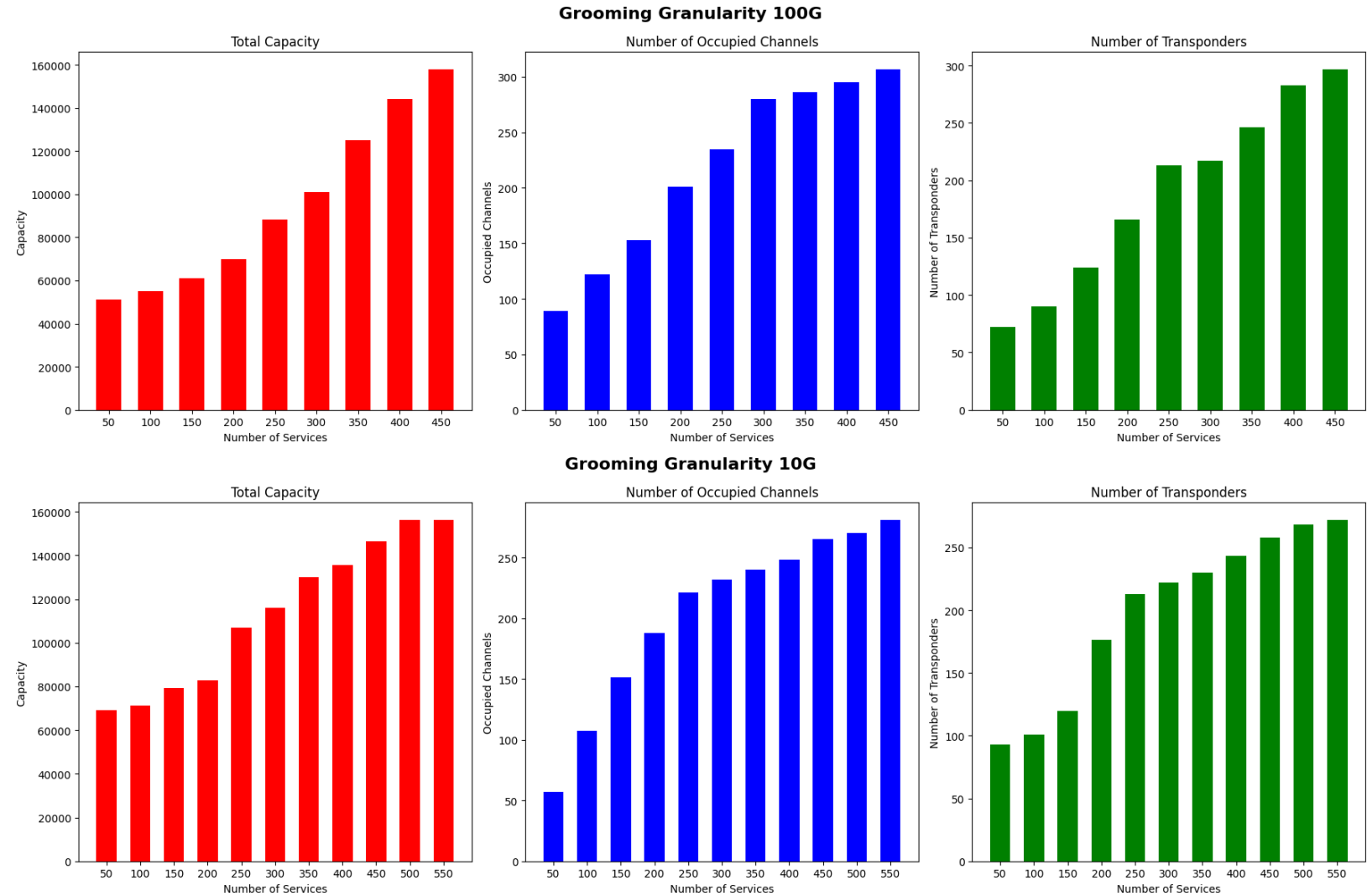


- National topology:
- #nodes=10
- #links=16



Results

- Continental topology:
- #nodes=14
- #links=24



National Graph	100G grooming	10G grooming
Number of services	362 (3.77)	453 (3.85)
Number of transponders	218 (1.70)	198 (1.13)
Number of occupied channels	162 (2.02)	134 (1.74)

Continental Graph	100G grooming	10G grooming
Number of services	471 (4.50)	557 (4.01)
Number of transponders	297 (2.43)	272 (1.77)
Number of occupied channels	307 (1.97)	281 (1.67)

Numbers in the table are mean value and numbers in the parenthesis are standard deviation

- Local search for improving heuristic routing of services
- Optimizing capacity and bandwidth utilization by choosing each OTN2 node grooming separately



Thank You

